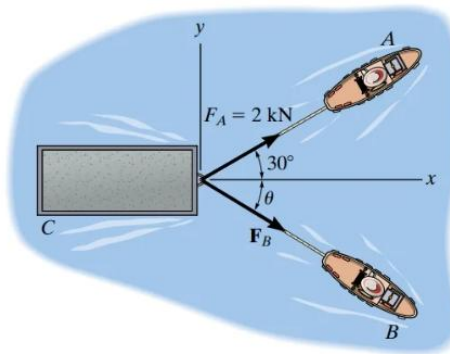


Oppgave 1: Find a system in which two or more forces are applied at a single point and take a picture of it (or get an image from the web). Generate a model of the system by making simplifications and reasonable assumptions about the magnitudes and directions of the forces, and answer the following questions:

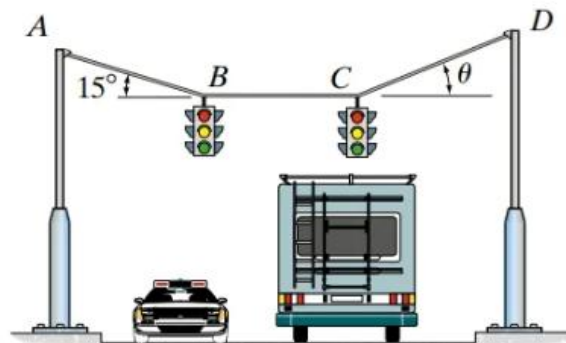
- Draw a diagram showing all forces as vectors.
- On that diagram, determine and plot the resultant force using the parallelogram law.
- Resolve each force into its X and Y components and calculate the magnitude and direction of the resultant force (you may need to rotate the x- and y- axes to simplify the analysis).

As an alternative, you may solve Exercise 2.28 of the textbook: “If the resultant force of the two tugboats is 3 kN, directed along the positive x axis, determine the required magnitude of force F_B and its direction θ ”.

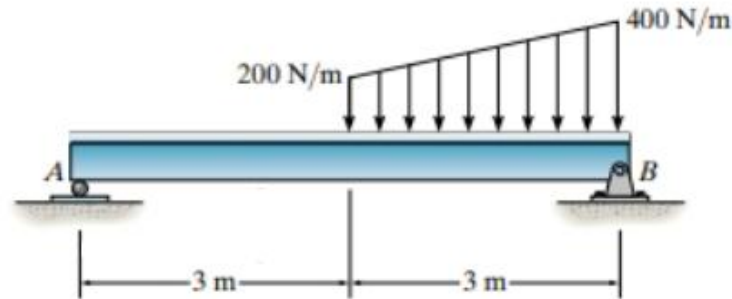


Oppgave 2: Find a system in which one or two objects (e.g., traffic lights, street lights, etc.) are suspended by cables. Develop a realistic model of the system by specifying the masses of the objects, relevant distances and angles, and any other necessary dimensions or assumptions. Then, calculate the tension in each cable. Clearly draw the free-body diagrams you have used and write the equations of equilibrium.

As an example, you may use the problem F3.6 of the textbook: “Determine the tension in cables AB, BC, and CD, necessary to support the 10-kg and 15-kg traffic lights at B and C, respectively. Also, find the angle θ ”.



Oppgave 3: Consider the beam with the distributed loading shown in the figure below. Draw the free-body diagram of the beam, including all external loads and support reactions. Replace the distributed load with two equivalent concentrated forces, indicating the magnitude and point of application of each force.



Next, find a real-world example of a beam subjected to a distributed load (the load distribution does not need to be the same as the one shown in the figure). Take a picture of your example (or obtain it from the web) and develop a simplified model by making appropriate assumptions about the geometry, loading, and support conditions so that the problem can be analysed using the principles of static.

Clearly state and justify your assumptions. Draw the free-body diagram, determine the equivalent force(s) associated with the distributed load, including its magnitude and point of application, and then calculate the support reactions. Show all calculations and equations of equilibrium used.

Oppgave 4: Select a real machine, mechanism, or frame structure composed of pin-connected members (e.g., a backhoe, forklift, scissor lift, excavator arm, bicycle frame, crane, engine hoist, or similar systems). Take a picture of the system (or obtain an image from the web) and develop a simplified statics model by making appropriate assumptions about the geometry, loading, member weights, and support conditions.

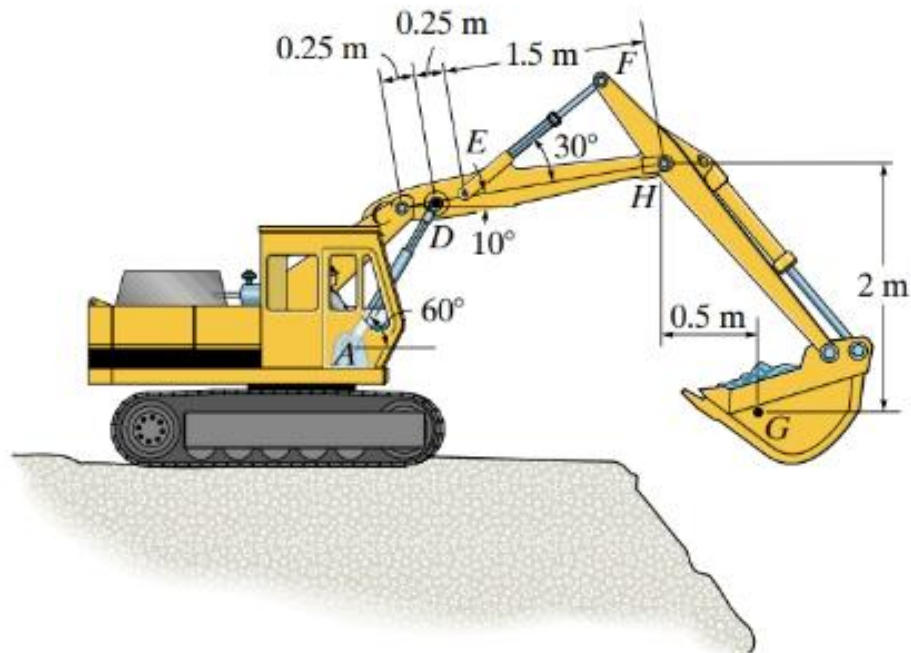
Assume that all members are rigid and pin connected. Clearly state and justify any additional assumptions needed for the analysis.

- Draw the free-body diagram of the entire system, showing all external forces and reactions.
- Draw the free-body diagrams of the individual components required for the analysis (e.g., members, hydraulic cylinders, pulleys, cables, buckets, loads, etc.).
- Identify a realistic external load acting on the system (e.g., the weight of a bucket and its contents, a lifted object, or an applied force) and estimate its magnitude.

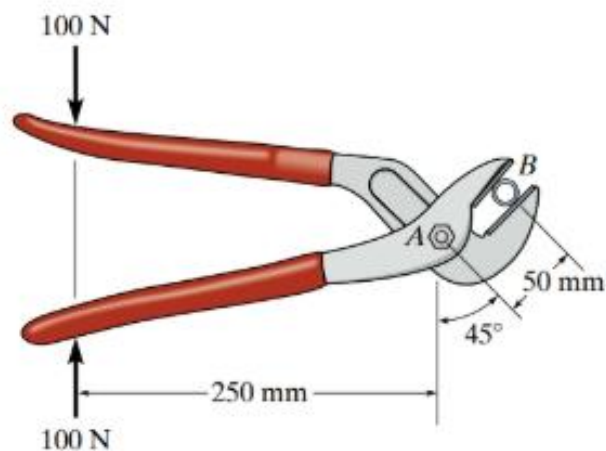
- d) Using the equations of equilibrium, determine the forces at the pin connections and in any two-force members, cables, or hydraulic cylinders that are part of the system. Show all calculations and clearly indicate the equilibrium equations used.

As examples, you may analyse the following systems

- Backhoe

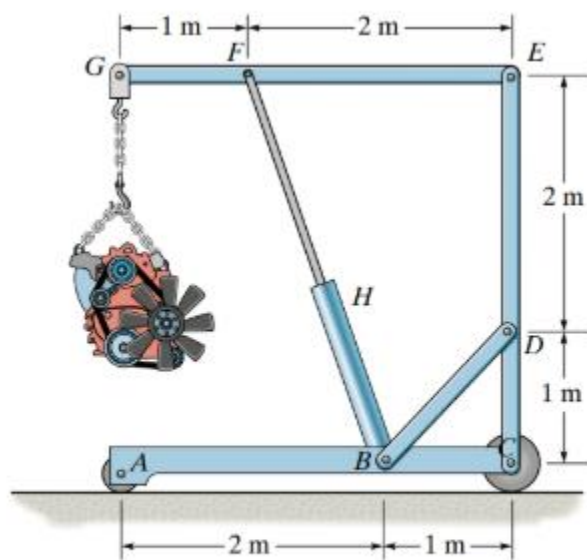


- Plier (exercise F6.15 – p172):



Prob. F6-15

- Hoist (exercise 6.82 -p179):



Prob. 6-82

Assessment criteria:

The exercises will be evaluated based on:

- 1) The appropriateness of the selected force systems.
- 2) The quality and justification of the modelling assumptions.
- 3) The correctness of the force diagrams and free-body diagrams.
- 4) The accuracy of the analytical calculations.
- 5) The clarity and organization of the work.

Notes:

- Choose systems that can reasonably be analysed using the methods covered in the course (i.e., statics). If a system is too complex, simplify it by making appropriate assumptions and clearly justify them. For example, a beam fixed at both ends has six unknown reaction components (F_x , F_y , and M at each support) but only three equilibrium equations can be used in this case. This makes the problem “statically indeterminate”, which cannot be solved using the principles of statics alone. In such cases, for example, you may assume different support conditions (e.g., a pin support and a roller support). Simpler, well-justified models are preferred over overly complex systems with unrealistic assumptions.
- In all exercises, clearly state and justify any assumptions made when modelling the systems.
- You may solve the examples proposed in each exercise. However, to obtain a grade of A or B, you are expected to analyse real-world cases rather than relying solely on the proposed examples.
- The originality of the examples you analyse will be positively considered, as will the use of real-world cases and realistic assumptions (e.g., weights, dimensions, material properties, and loading conditions).
- You may use AI tools for inspiration and to identify potential examples, but not to solve the exercises directly. Be aware that AI-generated content may contain errors, omissions, or hallucinations. You are responsible for the quality, accuracy, and integrity of the final work submitted.
- The use of AI should be acknowledged by describing how it was used (and which tools were used). Minor uses of AI tools (e.g., grammar or spelling checks) do not need to be reported.